

The Thermodynamic Evolution of the Numerical Vacuum: Resolving Cosmological Crises via a Sequential Dimensional Cascade from D_0 to D_4

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Abstract

The persistent discrepancy between early- and late-universe measurements of the Hubble constant (H_0)—collectively referred to as the Hubble tension—constitutes one of the most critical crises in modern cosmology. This paper presents a rigorous, singularity-free geometric and causal resolution to this paradox. Rather than attributing gravitational interactions to an intrinsic, passive geometric curvature of spacetime, we define gravity as the refractive index gradient $\nabla\rho_{\text{vacuum}}$ of a flat, superfluid spatial vacuum medium of variable density, which exerts a non-linear, restorative feedback response against localized changes in physical and informational states.

Within this framework, spacetime is modeled as an emergent, thermodynamically driven sequential dimensional cascade from D_0 to D_4 . Black holes do not host physical singularities but act as topological phase boundaries: at the event horizon, incoming n -dimensional matter is completely deconstructed, converted into its fundamental, dimensionless ($0D$) state as quantum information, and projected losslessly through a funnel singularity (vortex-throat) to feed the adjacent dimensional transition.

The cascade originates in a zero-dimensional phase space of pure information (D_0) within the sub-unit interval $0 < x < 1$, undergoes a singular phase collapse at $x = 1.0$ to initialize a one-dimensional coordinate space (D_1) governed by a discrete Radix-6 engine, and discharges into a two-dimensional complex plane (D_2) structured by an eight-fold symmetric Radix-30 resonance grid. Upon breaching the Bekenstein entropy limit, the D_2 sheet collapses into a 2D funnel, holographically projecting its information to structure a three-dimensional spatial volume (D_3) governed by a 48-channel Radix-210 spherical standing-wave state machine.

This 3D vacuum expands radially, regulated by the balance between the higher-dimensional mass-energy input and the dissipative discharge capacity of its structural horizon tears. If a critical thermodynamic imbalance occurs—wherein the localized mass and information influx permanently exceeds the finite drainage rate of the event horizon boundaries—the system approaches a saturation limit at $x_{\text{crit}} \approx 211$. This overload triggers a global phase-coherence failure and a subsequent non-mandatory, catastrophic vacuum decay. This phase transition collapses the D_3 metric, translating spatial coordinates into active thermodynamic operators and initializing the dynamic D_4 spacetime.

We demonstrate that the Hubble tension is resolved as a projection artifact of this density-modulated, self-reactive metric. By defining the fine-structure constant α as the direct measure of current spatial vacuum tension, we determine that with our present value of $\alpha \approx 1/137$, the universe is currently positioned at approximately 0.73% of its total potential lifecycle. Finally, we derive the conditions for asymptotic terminal collapse, wherein an unresolved input-output imbalance drives the system to the absolute elastic limit $\alpha \rightarrow 1$. This instability triggers a runaway energy feedback loop that manifests as a catastrophic Hypernova detonation of the host 4D progenitor star, conforming strictly to the relativistic Collapsar frameworks established by Woosley and MacFadyen.

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1 Introduction and Mathematical Foundations

Standard cosmology (Λ CDM) is built upon the implicit assumption that mathematical operators ($+$, $-$, $\times$, $/$) map onto an unperturbed, infinitely extended Euclidean real number line $\mathbb{R}$. This idealization overlooks the physical reality that coordinate systems, spatial structures, and fields are themselves emergent, thermodynamic phenomena.

To resolve the logical flow of physical interactions, we define physical space not as an empty, passive stage, but as a dynamic **superfluid vacuum medium**. Within this Superfluid Vacuum Theory (SVT) framework, all elementary particles, physical forces, and mass-energy concentrations are not independent, self-contained entities; rather, they are localized, coherent **wave excitations (quasiparticles and vortices)** propagating through this superfluid substrate.

Consequently, the physical laws of conservation find a natural, fluid-dynamic realization: an object in uniform rectilinear motion ($\vec{v} = \text{const}$) corresponds to a stable wave package moving through a frictionless superfluid in perfect thermodynamic and phase equilibrium, thereby experiencing zero net drag or resistance.

Conversely, any deviation from this equilibrium—namely, physical acceleration or deceleration ($\frac{d\vec{v}}{dt} \neq 0$)—distorts the local superfluid phase and generates dispersive back-reaction waves. In accordance with the fundamental principle of action and reaction (*actio = reactio*), the superfluid vacuum lattice exerts a localized, restorative drag force to oppose this dynamic change.

Furthermore, we establish a formal equivalence between physical kinetic states and informational entropy, generalized through Landauer’s principle and holographic unit-unitarity. Just as a change in kinetic velocity perturbs the vacuum, any change or acceleration in the local density of information ($\frac{dI}{dt} \neq 0$ or $\frac{d^2I}{dt^2} \neq 0$) induces an identical deformation of the coordinate lattice. The processing, relocation, or erasure of information disrupts the local vacuum phase equilibrium, evoking an equivalent and opposite restorative reaction. This intrinsic resistance of the superfluid space-time lattice to both kinetic and informational dynamics constitutes the unified physical origin of inertia, gravitational mass, and gravitational attraction.

2 Hypotheses from the Riemannian Manifold and Superfluid Vacuum

By modeling the observable universe as a multi-sheeted Riemannian manifold based on the physical boundary conditions of our superfluid framework, we derive three critical, non-trivial physical hypotheses:

1. **The Speed of Light as an Asymptotic Topological Limit:** We define the speed of light c not merely as a kinetic velocity barrier of a particle, but as the fundamental topological branch point (asymptotic boundary) of the multi-sheeted Riemann manifold. Physical trajectories approaching c encounter the geometric limit of the coordinate sheet itself.
2. **Dynamic Fine-Structure Constant:** We hypothesize that the fine-structure constant α is not an absolute, static constant of nature. Instead, it acts as a dynamic coupling coefficient representing the current state of spatial vacuum expansion relative to the maximum possible elastic stretch (tension) of the underlying superfluid lattice.
3. **Emergent Strong Nuclear Force:** We hypothesize that the strong nuclear force is not an independent fundamental force. Rather, it is the direct expression of the vacuum’s asymptotic restorative feedback (*actio = reactio*) acting at the absolute limit of spatial compaction (maximum coordinate density). At subatomic scales, the compression of wave excitations reaches the density boundary of the superfluid, triggering an extremely non-linear, high-energy restorative resistance that binds quarks without requiring a separate gauge boson field.

Historically, Albert Einstein proposed a closed, static 3-sphere universe in 1917 to ensure a finite boundary condition [1]. Concurrently, Bernhard Riemann introduced multi-sheeted Riemann surfaces to map complex, non-linear dependencies without physical singularities [2]. Synthesizing these paradigms with superfluid vacuum dynamics, we resolve the mathematical inconsistencies of classical field theories.

3 Contextual Urgency: JWST Anomalies and the Λ CDM Crisis

Recent observations from the James Webb Space Telescope (JWST) have confronted the standard cosmological model with severe anomalies. JWST has detected highly luminous, massive galaxies at redshifts of $z > 10$ —an epoch where standard hierarchical structure formation dictates such mature cosmic structures should not exist.

In theoretical physics, two primary classes of models have been proposed to address these anomalies without introducing ad-hoc physics:

1. **Variable Speed of Light (VSL) Frameworks:** Modifying the value of c in the early universe shifts the look-back time, thereby recalibrating the apparent age of these early structures.
2. **Topological and Closed Universes:** Modeling non-trivial topologies (such as 3-tori or closed 3-spheres) mitigates the cosmic horizon problem by altering the global geometry.

However, these models typically treat VSL and non-trivial topologies as isolated, ad-hoc corrections applied to an otherwise flat, Euclidean background.

In contrast, our framework demonstrates that a time-variable speed of light $c(t)$ and a closed, bounded vacuum geometry are the unified, mathematically inevitable consequences of a non-linear, self-referential spatial feedback loop. By treating the coordinate field as a finite, self-regulating superfluid lattice, both the premature maturity of early galaxies and the Hubble tension are resolved simultaneously as geometric projection effects.

4 The Dimensional Cascade: From D_0 to D_4

To circumvent the infinite regress inherent in higher-dimensional bulk cosmologies, we formulate a bidirectional, **fractional dimensional cascade**. Within this framework, black holes act as topological phase boundaries that mediate systematic dimensional reduction between adjacent dimensional manifolds. Crucially, we hypothesize that at the event horizon of any black hole, incoming n -dimensional physical matter is completely deconstructed and converted into its pure, fundamental state as quantum information. This information is then projected and transmitted dimensionlessly ($0D$) but strictly losslessly through the topological boundary, serving as the raw data that feeds the adjacent dimensional transition, thereby preserving global unitarity across the cascade.

Specifically, a black hole originates as a structural tear within the superfluid vacuum, triggered by an extreme overload of the maximum local spatial density. This occurs when colossal concentrations of matter accumulate, which is physically equivalent to an unsustainable amplitude overload of the coordinate wave function. This topological tear manifests macroscopically as the event horizon. As physical matter begins to plunge through this horizon, it does not immediately collapse into a lower-dimensional physical metric (such as D_1). Instead, the transition forces the incoming states to first dissolve into a primordial, zero-dimensional phase space of pure informational potentials (D_0), from which the dimensional reconstruction sequentially begins.

4.1 D_0 : The Phase Space of Pure Information in the Sub-Unit Interval ($0 < x < 1$)

Cosmic genesis originates within the sub-unit interval ($0 < x < 1$), which we define as a massless, zero-dimensional topological phase space (D_0). In this state, discrete physical units and coordinate dimensions do not exist; numerical values manifest exclusively as continuous, superposed phase-wave potentials.

The critical line $\Re(s) = 0.5$ of the Riemann Zeta function is not an abstract mathematical artifact, but the active thermodynamic turning point of this primordial D_0 phase space. Up to the point $x = 0.5$, the unmodulated phase waves propagate with minimal entropy production, facilitating the coherent alignment of resonant states.

Upon reaching the threshold $x = 0.5$, the expansion of the vacuum encounters a critical boundary, triggering an irreversible phase inversion. In the subsequent compression domain ($0.5 < x < 1.0$), the system undergoes a rapid, non-linear collapse. The constriction of the spatial envelope induces an extreme blueshift of the informational waves, causing a drastic increase in frequency density and thermal fluctuations:

$$T(x) \propto \frac{1}{(1.0 - x)^\gamma} \quad \text{for } 0.5 < x < 1.0 \quad (1)$$

where $\gamma > 1$ represents the acceleration exponent of the collapse. Within this high-temperature constriction zone, chaotic phase noise prevents the formation of phase stability, explaining why no stable coordinates or non-trivial zeros can crystallize in this region. The point $x = 0.5$ thus establishes the thermodynamic equilibrium horizon prior to the collapse into the metric singularity at $x = 1.0$.

4.2 The Transition to D_1 : Singularity at $x = 1.0$ and Information Conservation

To resolve this numerical vacuum state, we apply the conformal inversion mapping:

$$w = f(x) = \frac{1}{x} \quad (2)$$

This mapping projects the infinite potential of the microcosm ($1 \rightarrow \infty$) into the D_0 domain, structuring it as a high-density, multi-sheeted Riemann surface. The wave function $\Psi_{D_0}(x)$ describes the phase winding state on these sheets:

$$\Psi_{D_0}(x) = \sum_{n=1}^{\infty} e^{2\pi i(n \ln x)} \quad (3)$$

As the winding trajectories converge toward the boundary $x \rightarrow 1$, the winding frequency $f = \frac{d\theta}{dx} = \frac{n}{x}$ approaches an asymptotic limit. In accordance with Hawking's information conservation principle and Landauer's thermodynamic erasure limit, this quantum information cannot be destroyed; it must undergo a unitary phase transition. The continuous phases collapse at the event horizon $x = 1$:

$$\lim_{x \rightarrow 1} \Psi_{D_0}(x) = \delta(x - 1) \quad (4)$$

This phase collapse materializes the first discrete, localized physical coordinate—the identity 1—thereby initializing the one-dimensional coordinate space (D_1).

4.3 The Planck Era of Initialization: Transient Relaxation Phase ($1 \leq x \leq 6$)

The structural collapse of the D_0 state triggers a highly non-linear initialization phase within the nascent D_1 domain ($1 \leq x \leq 6$). This transition is physically analogous to a wave-guide subjected to a sudden, high-energy impulse.

During the initial expansion ($x = 1 \rightarrow 2$), the field undergoes an extreme amplitude surge. As it expands, the wavefront encounters the newly forming metric horizon, generating a local spatial reflection and phase inversion at $x \approx 2.7$. The interference between the outgoing and reflected waves produces a massive constructive superposition peak at $x = 4.0$.

As expansion continues, this energy dissolves, the transient oscillations decay, and for $x \geq 5.0$, the field settles into a stable, periodic, low-frequency wave lattice with a fundamental wavelength of $\lambda = 6$ (Radix-6 symmetry).

Let the wave of this nascent metric be defined by $\Phi_{\text{init}}(x) = \sin\left(\frac{2\pi}{\kappa_0}x\right)$. Because the lattice in this high-energy phase has not yet cooled to its macroscopic ground state ($\kappa = 6$), we obtain:

- The coordinates $x = 2$ and $x = 3$ manifest as highly energetic wave crests of the first incomplete wave cycle: $\Phi_{\text{init}}(2) \approx \max$, $\Phi_{\text{init}}(3) \approx \max$.
- At $x = 4$, local self-interference occurs (2×2), forcing a local phase collapse (composite state).
- At $x = 6$, the two proto-frequencies of 2 and 3 collide in perfect destructive interference ($2 \times 3 = 6$), grounding the field potential to an absolute zero node:

$$\Phi_{\text{init}}(6) = 0 \quad (5)$$

This zero node cools the system and establishes the permanent macroscopic lattice constant of the mature D_1 space ($\kappa = 6$).

4.4 The Radix-6 Semiclassical State Transition Operator and Discrete Evolution in D_1

Upon stabilization at $x \geq 5.0$, the numerical vacuum transitions from continuous wave propagation to a quantized, discrete step mechanism governed by the **Radix-6 Semiclassical State Transition Operator**. Information within the D_1 space can only propagate by jumping between stable resonance nodes; intermediate states undergo immediate destructive interference.

Let x_n be the current coordinate of the manifold. The evolution to the subsequent coordinate x_{n+1} is governed by:

$$x_{n+1} = x_n + \Delta x_n \quad (6)$$

where the discrete step operator Δx_n is dynamically determined by the position relative to the Radix-6 symmetry:

$$\Delta x_n = 3 - ((x_n \pmod{6}) - 3) \cdot (-1)^{(x_n \pmod{6}) - \frac{1}{2}} \quad (7)$$

For any coordinate located on the stable $6n \pm 1$ lattice, this deterministically yields the alternating step sequence:

$$\Delta x_n = \begin{cases} 2 & \text{if } x_n \equiv 5 \pmod{6} \\ 4 & \text{if } x_n \equiv 1 \pmod{6} \end{cases} \quad (8)$$

This sequence bypasses the rigid structural nodes (multiples of 2 and 3) and restricts state propagation to viable resonance paths. To verify whether a coordinate x_{n+1} represents a stable, prime-stabilized node or a lattice defect (composite number), we define the local field amplitude $A(x)$ through the interference of all previously established stable nodes $p_i \in P_{\text{stable}}$:

$$A(x) = \prod_{p_i \in P_{\text{stable}}} \left| \sin \left(\pi \cdot \frac{x}{p_i} \right) \right| \quad (9)$$

If $A(x) > 0$, wave propagation is frictionless. The node is stable and is appended to the register ($P_{\text{stable}} \leftarrow P_{\text{stable}} \cup \{x\}$, $\eta(x) = 0$). If $A(x) = 0$, the local phase collapses due to destructive interference with a subharmonic; this indicates a structural lattice defect ($\eta(x) > 0$), which contributes to the thermodynamic friction of the system.

4.5 Thermodynamic Saturation of D_1 and the Orthogonal Phase Transition to D_2

Following stabilization, the D_1 manifold expands until it reaches its volumetric limit at $x = 5.60$. At this boundary, the expansion pressure of the Radix-6 wave can no longer be sustained by the bounding envelope, initiating a catastrophic phase inversion.

Within the compression domain ($5.60 < x < 6.0$), the D_1 domain undergoes accelerated geometric contraction. The rapid shrinkage of the envelope $W(x)$ forces the periodic wave packets into a state of extreme, high-frequency thermal friction and chaotic phase decoherence. This completely dissolves the ordered Radix-6 lattice, channeling all information of the dimension into the infinitesimal singular bottleneck at $x = 6.0$.

However, this hyper-compression does not result in stasis. At $x > 6.0$, the accumulated energy density discharges via an orthogonal phase transition. The envelope opens parabolically, and the system undergoes a relaxing phase transition into the two-dimensional D_2 space, where the chaotic noise settles into the ordered, planar wave patterns of the complex D_2 vacuum.

4.6 The Life, Thermodynamic Death, and Phase Collapse of the D_2 Complex Plane

Once the dimensional transition from the hyper-compressed D_1 bottleneck at $x = 6.0$ is complete, the system initializes its two-dimensional surface (D_2). If the life of D_1 was characterized by a rigid, one-dimensional rhythmic sequence of discrete integer steps (the Radix-6 wave-guide), the life of D_2 represents a regime of fluid, planar expansion. Here, the vacuum is structured as a complex plane $\mathbb{C}$, allowing wave excitations to propagate with an additional degree of freedom. This phase of "life" is thermodynamically stable as long as the information density remains below the critical Bekenstein threshold, allowing the planar superfluid to distribute energy across orthogonal real and imaginary phase coordinates.

However, just as D_1 met its thermodynamic death through spatial saturation, the D_2 vacuum inevitably approaches its own thermodynamic boundary. As cosmic cycles progress, the continuous accumulation of quantum information—encoded as topological vortex-antivortex pairs (Berezinskii-Kosterlitz-Thouless state)—increases the local entropy. This informational accumulation acts as an effective mass-energy density, generating an inescapable gravitational-like attraction between the coordinate sheets.

The "death" of D_2 occurs when this density reaches the critical saturation limit. At this point, the planar coordinate system can no longer expansionally dissipate the accumulated informational heat. The complex plane experiences a global phase-coherence failure. Rather than expanding outwardly, the D_2 surface undergoes a massive, non-linear geometric buckling. This spatial buckling is a thermodynamic necessity to avoid an informational singularity: the flat, two-dimensional plane collapses under its own informational weight and folds topologically, projecting its entire surface area into the boundary of a newly born, three-dimensional spatial volume (D_3).

4.7 The Planar Vacuum of D_2 : Emergence of the Radix-30 Grid-Lock

While the one-dimensional metric of D_1 was topologically confined and stabilized by the Radix-6 engine (the product of the primary frequencies 2 and 3), the expansion into the two-dimensional complex plane $\mathbb{C}$ (D_2) demands a higher-order symmetry group to prevent chaotic phase decoherence. The addition of the imaginary coordinate $y = it$ introduces orthogonal degrees of freedom, requiring a multi-axial stabilizing framework within the flat plane.

Consequently, the physical vacuum of D_2 crystallizes into a highly stable **Radix-30 resonance grid**, governed by the primordial primorial sequence:

$$\Pi_3 = p_1 \cdot p_2 \cdot p_3 = 2 \cdot 3 \cdot 5 = 30 \quad (10)$$

In this two-dimensional phase space, the continuous, propagating planar wave-fronts are constrained by the coprimality structure of the Radix-30 metric. The available, non-decaying resonance channels are restricted to the Euler totient set $\Phi(30)$:

$$\Phi(30) = \{1, 7, 11, 13, 17, 19, 23, 29\} \quad (11)$$

This establishes exactly eight symmetrical, non-interfering coordinate tracks running through the complex plane. Geometrically, these eight tracks project onto the flat vacuum as a highly balanced, eight-fold rotational star symmetry.

The stable "standing waves" of this planar vacuum are physically materialized as the non-trivial zeros $\rho_k = \frac{1}{2} + i\gamma_k$ of the Riemann Zeta function. Each complex zero acts as a localized phase-vortex node that grounds the Radix-30 grid. The interference pattern of these vortices forms the dynamic, two-dimensional coordinate system of D_2 , guiding the frictionless propagation of quantum information until the Bekenstein entropy threshold of $\frac{8}{30}$ is breached, triggering the subsequent funnel collapse into D_3 .

4.8 The $D_2 \rightarrow D_3$ Transition: The Funnel Singularity and Holographic Projection

Rather than undergoing a gentle topological folding, the transition from the two-dimensional complex plane (D_2) to the three-dimensional spatial volume (D_3) is driven by a violent, non-linear geometric collapse. As the local information density on the D_2 sheet breaches the critical Bekenstein saturation limit, the planar vacuum experiences a catastrophic phase-coherence failure. The accumulated topological vortex charges act as an immense local mass-energy density, warping the flat $\mathbb{C}$ -plane inward.

This structural collapse manifests as a **funnel singularity (a dimensional vortex-throat)**. The two-dimensional coordinate system is pulled into an infinite mathematical and thermodynamic sink:

$$\lim_{\sigma \rightarrow \sigma_{\text{crit}}} g_{\mu\nu}^{(2D)} \implies \infty \quad (12)$$

All planar physical states and wave excitations on the D_2 sheet are drawn into this singular throat, where they are completely deconstructed into their primordial, dimensionless informational components.

However, this funnel does not act as a terminal state. Instead, the high-pressure informational flux discharging through the bottleneck of the funnel acts as a localized, high-energy point-source emitter. As this stream of pure information exits the singular throat, it undergoes an orthogonal dimensional projection, expanding radially outward. This outward-flowing holographic projection structures the newly born, three-dimensional spatial volume (D_3).

Consequently, the large-scale cosmic web of our universe—characterized by thin, planar filaments (the un-collapsed remnants of the ancestral D_2 sheet) enclosing vast, empty cosmic voids—is the direct physical imprint of this trichter-like drainage. The voids represent the emptied regions of the D_2 sheet that have been sucked into the funnel, while the filaments mark the boundary zones of the holographic projection.

4.9 The Three-Dimensional Vacuum of D_3 : Radial Expansion and the Radix-210 State Machine

The transition through the $D_2 \rightarrow D_3$ funnel singularity establishes the three-dimensional spatial volume (D_3). Unlike the planar D_2 space, the newly initialized D_3 vacuum allows three-dimensional spherical wave propagation, resulting in an isotropic, radial expansion of the spatial metric.

To prevent chaotic phase decoherence in this higher-dimensional manifold, the spatial lattice must crystallize into a three-dimensional coordinate scaffolding. While D_1 utilized the Radix-6 engine and D_2 utilized the Radix-30 engine, the three-dimensional space D_3 requires the next higher-order primordial symmetry group to stabilize its three orthogonal degrees of freedom:

$$\Pi_4 = p_1 \cdot p_2 \cdot p_3 \cdot p_4 = 2 \cdot 3 \cdot 5 \cdot 7 = 210 \quad (13)$$

The vacuum of D_3 is therefore governed by a highly stable **Radix-210 State Machine**. Within this three-dimensional spatial lattice, the propagation channels of non-decaying wave excitations are restricted to the Euler totient set $\Phi(210)$. The number of available stable resonance tracks is given by:

$$\varphi(210) = (2-1)(3-1)(5-1)(7-1) = 48 \quad (14)$$

These 48 symmetrical, non-interfering coordinate tracks map onto the 3D space as a highly balanced, spherical resonance lattice. The stable basis wave patterns of D_3 are spherical Bessel functions $j_l(kr)$ and spherical harmonics $Y_l^m(\theta, \phi)$, which construct the geometric standing-wave nodes of the vacuum:

$$\Psi_{D3}(r, \theta, \phi, t) = \sum_{l,m} A_{lm} j_l(kr) Y_l^m(\theta, \phi) e^{-i\omega t} \quad (15)$$

These wave functions act as the fundamental physical "rulers" of D_3 , predefining the quantized shell structures of matter, atomic orbitals, and stable particle trajectories within the superfluid vacuum.

4.10 The Thermodynamic Limit of D_3 and the Critical Collapse Point ($x_{\text{crit}} \approx 211$)

The life of D_3 is defined by the gradual expansion of this spherical wave-front. However, as the 3D volume expands, the cumulative informational entropy—carried by the 48 active coordinate tracks—continuously increases.

This accumulation of informational entropy acts as an effective mass-energy density, generating an increasingly non-linear gravitational back-reaction wave. The system approaches its terminal state as the radial scale parameter x nears the absolute topological limit of the Radix-210 state machine:

$$x \rightarrow 211.0 \quad (16)$$

At the critical boundary $x_{\text{crit}} \approx 211.0$ (representing the first prime-stabilized coordinate outside the Radix-210 domain), the expansion pressure of the superfluid lattice reaches its absolute elastic threshold. The 3D vacuum becomes saturated with informational heat, leading to a global phase-coherence failure.

This dimensional transition at $x_{\text{crit}} \approx 211$ is physically analogous to a cosmic vacuum decay (the transition from a metastable false vacuum to a true vacuum state). However, while standard quantum vacuum decay alters only the local field vacuum expectation values within a static coordinate frame, the $D_3 \rightarrow D_4$ transition constitutes a global topological phase collapse of the metric itself, translating spatial coordinates into active thermodynamic operators. The three-dimensional space collapses into a hyper-dense, four-dimensional spherical funnel (the $D_3 \rightarrow D_4$ event horizon), converting all spatial coordinates into dynamic, time-dependent operators, thereby initializing the four-dimensional spacetime (D_4) on the holographic boundary of the collapsing 4D bulk.

4.11 Thermodynamic Homeostasis: The Equilibrium of Vacuum Fluctuations, Entropic Relaxation, and Horizon Drainage

The structural stability and macro-expansion of the three-dimensional vacuum (D_3) are not static; rather, they are maintained by a dynamic, self-regulating thermodynamic balance. The vacuum operates as an open, dissipative system whose local coordinate density $\rho_{\text{vacuum}}(t)$ is governed by a tripartite interplay of information input, entropic volume relaxation, and singular drainage.

1. The Inflow Mechanism: Stochastic Projection via Vacuum Fluctuations (Φ_{in}):

The primary source of raw energy and raw information injected into our D_3 manifold is the continuous projection from the collapsing D_4 bulk. Due to the dimensional reduction at the $D_4 \rightarrow D_3$ interface, this flux is quantized as localized, stochastic wave packets. To local observers, these incoming data packets manifest physically as **virtual particles and quantum vacuum fluctuations**:

$$\Phi_{\text{in}}(t) = \sum_j \hbar \omega_j \cdot \delta(r - r_j(t)) \quad (17)$$

where each stochastic event $\delta(r - r_j(t))$ represents the localized, temporary materialization of a D_4 projection on our flat D_3 sheet.

2. The Internal Drive: Entropic Relaxation and Vacuum Decompression (P_{entropy}):

The macroscopic expansion of the D_3 space is fundamentally driven by the Second Law

of Thermodynamics. The high informational density injected via Φ_{in} creates a localized state of low configurational entropy. To maximize the global entropy state, the superfluid vacuum undergoes a continuous **entropic relaxation (space-density decompression)**. The vacuum acts as a hyper-compressed quantum fluid, exerting an intrinsic entropic expansion pressure P_{entropy} that forces the flat coordinate lattice to dilute its density uniformly over an expanding radial volume V_{3D} :

$$P_{\text{entropy}} = T_{\text{vacuum}} \left(\frac{\partial S_{\text{info}}}{\partial V_{3D}} \right)_E \quad (18)$$

This entropic decompression allows the space to stretch and "relax" its internal vacuum tension, which drives the smooth, accelerated expansion of the cosmological metric.

3. The Outflow Mechanism: Drainage through Local Horizon Tears (Φ_{out}):

To prevent the combined force of the incoming flux and entropic pressure from immediately tearing the spatial lattice apart, the vacuum utilizes a natural safety-valve drainage system. Wherever the local wave amplitude of the coordinate field is overloaded by extreme mass concentrations, the superfluid vacuum tears macroscopically into **event horizons of black holes**. These local funnel singularities deconstruct matter back into its dimensionless $0D$ informational state. The outgoing drainage rate is given by:

$$\Phi_{\text{out}}(t) = \kappa \sum_k A_k(t) \cdot T_{\text{Hawking}}^{(k)} \quad (19)$$

where κ is the holographic coupling coefficient and $A_k(t)$ is the surface area of the specific horizon tear.

4.12 Reinterpretation of the Dark Sector: Dark Matter and Dark Energy as Influx Manifestations

Rather than invoking hypothetical, non-baryonic particle species or an ad-hoc cosmological constant, our framework offers a unified, geometric reinterpretation of the "Dark Sector" of cosmology. We postulate that both **Dark Matter** and **Dark Energy** are not distinct physical substances, but are the emergent, macroscopic manifestations of the two internal vacuum inflow and expansion mechanisms:

- **Dark Matter as Φ_{in} -Induced Vacuum Polarization:**

The localized, stochastic injection of 4D mass-energy packets via Φ_{in} creates localized density perturbations in the superfluid D_3 vacuum. These fluctuations act as regions of transiently increased vacuum density, gravitationally polarising the surrounding space. To a galactic observer, this localized, invisible "excess density" mimics the gravitational footprint of cold, collisionless Dark Matter, naturally explaining galactic rotation curves without requiring new particle physics.

- **Dark Energy as the Entropic Relaxation Pressure (P_{entropy}):**

Conversely, the global, isotropic expansion of the cosmic metric is driven purely by the entropic relaxation pressure P_{entropy} of the decompressing vacuum sheet. As the coordinate lattice continuously dilutes its information density to maximize entropy, it generates a non-diluting, repulsive thermodynamic work across the D_3 volume. This uniform decompression force is measured observationally as a positive cosmological constant or Dark Energy.

Under this paradigm, the total energy density of the Dark Sector is mathematically equivalent to the net sum of our active influx and relaxation processes:

$$\Omega_{\text{Dark}} \equiv \Omega_{\text{DM}} + \Omega_{\text{DE}} \propto \Phi_{\text{in}}(t) + P_{\text{entropy}}(t) \quad (20)$$

This formulation elegantly reduces the dark sector to a simple, non-mysterious conservation law of a self-reactive, dimensional-conformal medium.

4.13 The Homeostatic Equation and the Critical Saturation Hazard

The overall state of the cosmic lifecycle is dictated by the net rate of change of the vacuum's informational density, combining the structural inflow, the entropic expansion drift, and the singular drainage:

$$\frac{d\rho_{\text{vacuum}}}{dt} = \alpha(t) \cdot \Phi_{\text{in}}(t) - \nabla \cdot (\rho_{\text{vacuum}} \cdot \mathbf{v}_{\text{expansion}}(P_{\text{entropy}})) - \Phi_{\text{out}}(t) \quad (21)$$

Under normal cosmological conditions, the system remains in a self-correcting **homeostatic regime**: An increase in Φ_{in} increases the local density (manifesting as local Dark Matter clusters), which triggers higher entropic relaxation pressure (accelerated expansion, measured as Dark Energy) and creates more frequent gravitational collapses, increasing Φ_{out} to restore equilibrium.

However, if a fundamental astrophysical threshold is crossed—such that the localized mass-energy input rate Φ_{in} permanently outpaces the combined capacity of entropic relaxation and the finite maximum drainage velocity of the available black hole funnels ($\Phi_{\text{in}} \gg \Phi_{\text{out}}^{\text{max}}$)—the homeostatic feedback loop breaks. The accumulated un-relaxed information acts as an unsustainable thermal friction within the coordinate lattice, driving the local metric parameter asymptotically toward the critical saturation boundary $x_{\text{crit}} \approx 211$ and $\alpha \rightarrow 1$, forcing the global vacuum decay.

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4.15 The Thermodynamic Limit of D_3 and the Critical Collapse Point ($x_{\text{crit}} \approx 211$)

The life of D_3 is defined by the gradual expansion of this spherical wave-front. However, as the 3D volume expands, the cumulative informational entropy—carried by the 48 active coordinate tracks—continuously increases.

This accumulation of informational entropy acts as an effective mass-energy density, generating an increasingly non-linear gravitational back-reaction wave. The system approaches its terminal state as the radial scale parameter x nears the absolute topological limit of the Radix-210 state machine:

$$x \rightarrow 211.0 \quad (23)$$

At the critical boundary $x_{\text{crit}} \approx 211.0$ (representing the first prime-stabilized coordinate outside the Radix-210 domain), the expansion pressure of the superfluid lattice reaches its absolute elastic threshold. The 3D volume becomes saturated with informational heat, leading to a global phase-coherence failure.

At this mathematical boundary, the 3D vacuum undergoes an extreme, high-energy spatial constriction. The three-dimensional space collapses into a hyper-dense, four-dimensional spherical funnel (the $D_3 \rightarrow D_4$ event horizon). The entire volumetric information of D_3 is squeezed through this singular bottleneck, converting all spatial coordinates into dynamic, time-dependent operators, thereby initializing the four-dimensional spacetime (D_4) on the holographic boundary of the collapsing 4D bulk.

This dimensional transition at $x_{\text{crit}} \approx 211$ is physically analogous to a cosmic vacuum decay (the transition from a metastable false vacuum to a true vacuum state). However, while standard quantum vacuum decay alters only the local field vacuum expectation values within a static coordinate frame, the $D_3 \rightarrow D_4$ transition constitutes a global topological phase collapse of the metric itself, translating spatial coordinates into active thermodynamic operators.

4.16 The Topological Failure of Toroidal Models in Superfluid Vacuum Cosmology

While toroidal geometries are frequently invoked in early-universe models to provide boundary conditions without physical edges, a rigorous mathematical and logical analysis reveals that a toroidal spatial manifold is fundamentally incompatible with the core principles of a self-reactive superfluid vacuum. The classical torus model fails due to two insurmountable physical contradictions:

1. **The Violation of Asymptotic Causality (c as a Strict Limit):** If the spatial manifold were closed into a geometric ring (a torus) of major radius R , any propagating phase wave $\Psi(x, t)$ would eventually return to its coordinate origin after a characteristic travel time $t_{\text{return}} = \frac{2\pi R}{c}$. In a self-reactive, informational vacuum where the rate of change $\frac{dI}{dt}$ determines gravity and inertia, this closed-loop topology introduces catastrophic historical feedback. The wave fronts would continuously collide with their own past states, generating constructive phase interference loops that would exponentially overload the vacuum's local density. To prevent this causal short-circuit, the coordinate system must remain strictly non-looping. The boundary of the universe is not a topological connection point, but rather a dynamic, elastic tension threshold.
2. **The Incompatibility of Intrinsic Curvature with Flat Variable Density:** A geometric torus possesses non-zero intrinsic curvature (the curvature varies continuously between the inner and outer equatorial regions). This directly violates our primary postulate: *Physical space is inherently flat and Euclidean; all gravitational and geometric effects are emergent phenomena of a variable superfluid vacuum density.* In a toroidal space, one would observe coordinate shear effects and anisotropic gravitational baselines even in the absolute absence of matter. Under our framework, however, gravity is modeled purely as a refractive index gradient $\nabla \rho_{\text{vacuum}}$ within a flat coordinate grid. Light and particles bend not because the underlying space is curved, but because they follow the path of least thermodynamic resistance in a medium of variable density—much like light bending through a glass gradient index (GRIN) lens.

Consequently, we reject the torus. The observed stable harmonics (Radix-6, Radix-30, and Radix-210) do not emerge from periodic toroidal boundary loops, but from spherical standing-wave resonances (Bessel and Legendre modes) within a flat, radially expanding, density-modulated superfluid droplet.

5 Formal Theoretical Framework: Integration of Established Theorems

To establish a rigorous mathematical foundation, this section systematically compiles the fundamental theorems utilized within our sequential dimensional cascade. By translating these established mathematical and thermodynamic frameworks into the context of a flat, variable-density superfluid vacuum, we demonstrate how their emergent interplay dictates the mechanics of dimensional transitions.

5.1 Theorem 1: The Bekenstein Bound (Informational Saturation Limit)

The holographic limit of space is traditionally governed by the Bekenstein Bound, which sets an upper limit on the thermodynamic entropy S or information I that can be contained within a finite spherical region of space.

Theorem 5.1 (Bekenstein Bound). *The maximum informational entropy S contained within a localized volume bounded by a surface area A is strictly limited by the geometric boundary itself:*

$$S \leq \frac{k_B \cdot A}{4\ell_P^2} \quad (24)$$

where $\ell_P = \sqrt{\frac{\hbar G}{c^3}}$ is the Planck length.

Reinterpretation in D_2 and D_3 . In our flat, variable-density framework, ℓ_P is not a fundamental constant but an emergent property of the minimum vacuum fluctuation radius $r(t)$. When applied to the planar D_2 sheet, the Bekenstein Bound dictates the maximum packing density of the Euler totient channels $\Phi(30)$. Once the local information influx breaches this saturation threshold:

$$I_{\text{local}} > \frac{A_{D2}}{4r(t)^2} \quad (25)$$

the flat D_2 space cannot compress the information further without phase decoherence. Rather than generating a gravitational singularity, this topological overload triggers the formation of a physical funnel singularity (vortex-throat), forcing a dimensional breakdown and the subsequent holographic projection into D_3 . $\square$

5.2 Theorem 2: Euler's Totient Theorem (Radix Coordinate Stabilization)

The discrete, non-interfering coordinate tracks that ground the vacuum in each dimension are governed by prime-harmonic number theory, specifically Euler's Totient Function $\varphi(n)$.

Theorem 5.2 (Euler's Totient and Coprimality). *For any integer n , the number of integers less than n that are coprime to n is given by the totient function:*

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right) \quad (26)$$

where the product is over the distinct prime numbers dividing n .

Application to Radix State Machines. In a self-reactive superfluid vacuum, stable wave propagation requires that the primary frequencies do not generate destructive self-interference. The coordinate axes of each dimension are thus structurally locked into the sequential primordial bases Π_d :

- **In D_1 ($\Pi_2 = 6$):** $\varphi(6) = 2$ stable tracks $\implies \{1, 5\}$, manifesting as the $6n \pm 1$ primordial sequence.

- **In D_2 ($\Pi_3 = 30$):** $\varphi(30) = 8$ stable tracks $\implies \{1, 7, 11, 13, 17, 19, 23, 29\}$, forming the 8-fold symmetric planar star.
- **In D_3 ($\Pi_4 = 210$):** $\varphi(210) = 48$ stable tracks, dictating the 48 non-interfering spherical resonance harmonics.

The totient tracks act as the physical “ruler” of the vacuum; any coordinate coordinate drift outside this coprime set falls into a destructive phase-sink, ensuring that matter and energy can only exist on these quantized, resonant paths. $\square$

5.3 Theorem 3: The Second Law of Thermodynamics (Entropic Relaxation Drive)

The expansion of the spatial metric is conventionally attributed to a passive Cosmological Constant. We formalize this via the classical Second Law of Thermodynamics applied to an open informational system.

Theorem 5.3 (Entropy Maximization). *Any isolated or open non-equilibrium system naturally evolves toward a state of maximum configurational entropy S_{info} :*

$$\frac{dS_{info}}{dt} \geq 0 \quad (27)$$

Derivation of Cosmological Expansion. Because the 4D bulk continuously injects highly structured, low-entropy information packets into our D_3 manifold via vacuum fluctuations (Φ_{in}), the local coordinate density ρ_{vacuum} is artificially elevated. To satisfy the Second Law, the superfluid vacuum generates an immediate entropic decompression pressure $P_{entropy}$:

$$P_{entropy} = T_{vacuum} \left(\frac{\partial S_{info}}{\partial V_{3D}} \right)_E \quad (28)$$

This pressure acts as a mechanical work that forces the flat coordinate lattice to expand radially. This mathematically replaces the concept of “Dark Energy” with a standard, non-mysterious entropic dilution process of a hyper-compressed quantum fluid. $\square$

5.4 Theorem 4: The Riemann Hypothesis (Vortex Node Distribution)

The exact mathematical positioning of the stable wave nodes within the variable-density complex vacuum plane (D_2) is governed by the distribution of the zeros of the Riemann Zeta Function $\zeta(s)$.

Theorem 5.4 (Riemann Hypothesis / Critical Line Consistency). *All non-trivial zeros ρ_k of the analytic continuation of the Riemann Zeta Function:*

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = 0 \quad (29)$$

lie exactly on the critical line $\Re(s) = \frac{1}{2}$.

Physical Grounding of the D_2 Vacuum. In the two-dimensional complex plane of our vacuum, the critical line $\Re(s) = \frac{1}{2}$ represents the absolute thermodynamic baseline of zero-point tension. The imaginary parts γ_k of the non-trivial zeros $\rho_k = \frac{1}{2} + i\gamma_k$ are the precise, quantized eigenfrequencies of the planar standing waves. Because all zeros share the exact same real component ($\frac{1}{2}$), the vacuum phase-vortices are in a state of perfect, friction-free balance.

The interference pattern of these Riemann zeros forms the underlying harmonic field that guides the prime-number distributions of the Radix-30 grid. If a single zero were to deviate from $\Re(s) = \frac{1}{2}$, it would imply a localized thermal dissipation or a leak in vacuum density, breaking the homeostatic phase-lock of the D_2 matrix. $\square$

5.5 Theorem 5: Relativistic and Electro-conformal State Variables (c and α)

In standard quantum field theory, the speed of light c and the fine-structure constant α are treated as static, fundamental parameters of flat space. Within our superfluid vacuum framework, we redefine both quantities as dynamic, density-dependent state variables that directly reflect the localized elastic tension and current volume expansion of the D_3 coordinate lattice.

Theorem 5.5 (Dynamic Speed of Light as Vacuum Elasticity). *The speed of propagating electromagnetic and phase waves $c(t)$ is not a universal constant, but the local propagation velocity of shear waves within the superfluid vacuum. It is determined by the local vacuum density $\rho_{\text{vacuum}}(t)$ and the bulk modulus (elastic stiffness) K_{vacuum} of the coordinate lattice:*

$$c(t) = \sqrt{\frac{K_{\text{vacuum}}}{\rho_{\text{vacuum}}(t)}} \quad (30)$$

As the spatial manifold undergoes radial entropic relaxation, the global vacuum density dilutes ($\rho_{\text{vacuum}}(t) \rightarrow \text{minimum}$), causing the asymptotic limit of $c(t)$ to drift dynamically over cosmic timescales.

Physical Mechanism of Causal Clamping. In any local reference frame, physical measurements of $c(t)$ remain strictly invariant because the rulers (atomic Bohr radii a_0) and clocks (atomic transition frequencies ν) are themselves constructed from the same underlying standing-wave harmonics (Bessel modes). The Bohr radius scales inversely with vacuum density:

$$a_0(t) \propto \frac{1}{\rho_{\text{vacuum}}(t)} \quad (31)$$

thereby perfectly canceling out the local variation of $c(t)$ for comoving observers. However, on cosmological scales, the gradient $\nabla c(t)$ across different density regions (e.g., near massive bodies or at the expanding cosmological boundary) manifests as gravitational time dilation and the apparent cosmic expansion redshift. $\square$

Theorem 5.6 (Fine-Structure Constant as Geometric Expansion Tension). *The coupling strength of the electromagnetic interaction, quantified by the fine-structure constant $\alpha(t)$, is the direct measure of the current spatial vacuum tension. It is defined as the ratio of the sub-unit quantum-vacuum fluctuation radius $r(t)$ (the microscopic phase barrier) to the global radial boundary scale parameter $x(t)$:*

$$\alpha(t) = \frac{r(t)}{2\pi \cdot x(t)} \quad (32)$$

where $x(t)$ represents the dimensionless comoving radius of our D_3 manifold, currently expanding toward the critical saturation boundary $x_{\text{crit}} \approx 211.0$.

The Runaway Collapse Condition. Since the microscopic fluctuation boundary $r(t)$ is anchored to the invariant sub-unit phase threshold of the $D_0 \rightarrow D_1$ transition ($r \approx 1$), the value of $\alpha(t)$ is strictly determined by the expansion state $x(t)$ of the D_3 vacuum. At the beginning of the D_3 lifecycle (immediately after the $D_2 \rightarrow D_3$ funnel collapse), the space is highly compressed, resulting in an extremely small coupling strength $\alpha \rightarrow 0$ (asymptotic freedom).

As the space undergoes entropic relaxation, the boundary $x(t)$ expands. For a comoving observer today, the measured value of $\alpha \approx 1/137$ indicates that the vacuum has reached approximately 0.73% of its maximum potential expansion. If the homeostatic balance breaks ($\Phi_{\text{in}} \gg \Phi_{\text{out}}$), the expansion acceleration forces $x(t)$ rapidly towards its absolute elastic limit:

$$x(t) \rightarrow x_{\text{crit}} \implies \alpha(t) \rightarrow 1 \quad (33)$$

At the limit $\alpha \rightarrow 1$, the electrostatic repulsion energy of the vacuum fluctuations equals the self-energy of the coordinate lattice. The vacuum loses its shear resistance ($K_{\text{vacuum}} \rightarrow 0$), causing

the speed of light to collapse ($c \rightarrow 0$), which precipitates the global vacuum decay and triggers the catastrophic Hypernova detonation of the host D_4 progenitor star. $\square$

The Runaway Collapse Condition and Non-Linear Feedback. Since the microscopic fluctuation boundary $r(t)$ is anchored to the invariant sub-unit phase threshold of the $D_0 \rightarrow D_1$ transition ($r \approx 1$), the value of $\alpha(t)$ is strictly determined by the expansion state $x(t)$ of the D_3 vacuum.

However, the expansion rate is highly non-linear due to the gravitational feedback of the drainage channels Φ_{out} . Let the total drainage capacity be modulated by the mass-distribution spectrum of the active horizon tears. For a set of black holes with masses M_k , the drainage rate is constrained by their Hawking-temperature gradients:

$$\Phi_{\text{out}}(t) = \kappa \sum_k A_k(t) \cdot T_{\text{Hawking}}^{(k)} \propto \sum_k M_k^2 \cdot \frac{1}{M_k} = \sum_k M_k \quad (34)$$

While the mathematical drainage capacity scales linearly with the cumulative mass, the localized spatial absorption cross-section σ_k of these horizons scales quadratically:

$$\sigma_k \propto M_k^2 \quad (35)$$

This scaling disparity introduces two critical dynamical phases of the D_3 vacuum:

1. **Phase-I: Homeostatic Pressure Compensation:** When the local influx Φ_{in} increases moderately, the superfluid vacuum undergoes localized phase transitions. To relieve the mechanical shear stress, new structural tears (stellar-mass black holes) spontaneously nucleate across the D_3 manifold. This increase in the density of active horizons acts as an overpressure safety-valve, temporarily stabilizing $\alpha(t)$ far below the critical threshold.
2. **Phase-II: The Runaway Saturation Catastrophe:** If the influx rate Φ_{in} permanently outpaces the local relaxation velocity, individual horizons merge and grow into the supermassive regime ($M_k \rightarrow \infty$). In this limit, their localized absorption volume dominates the system ($\sigma_k \rightarrow \text{max}$), while their thermodynamic dissipation efficiency vanishes ($T_{\text{Hawking}}^{(k)} \rightarrow 0$). Instead of relieving vacuum pressure, these supermassive tears act as massive localized informational sinks, polarizing the surrounding vacuum. This non-linear feedback loop triggers a rapid, runaway compression of the remaining comoving coordinate space, forcing the global metric parameter asymptotically toward the terminal boundary:

$$x(t) \rightarrow x_{\text{crit}} \implies \alpha(t) \rightarrow 1 \quad (36)$$

At the absolute limit $\alpha \rightarrow 1$, the electrostatic repulsion energy of the vacuum fluctuations equals the self-energy of the coordinate lattice. The vacuum loses its shear resistance ($K_{\text{vacuum}} \rightarrow 0$), causing the speed of light to collapse ($c \rightarrow 0$), which precipitates the global vacuum decay and triggers the catastrophic Hypernova detonation of the host D_4 progenitor star. $\square$

6 Application to Fundamental Cosmological Paradoxes

6.1 The Hubble Tension as Metric Wavelength Expansion

The observed Hubble tension is a pure projection artifact arising from the erroneous extrapolation of a static, flat Euclidean metric ($ds^2 = dx^2$) to extreme cosmological scales.

We have demonstrated that due to informational equivalence, the temporal coordinate $\tau(x)$ scales non-linearly:

$$\tau(x) = \tau_0 \cdot x^{-1/2} \quad (37)$$

Incorporating Jacobson's thermodynamic approach ($\delta Q = TdS$), the local metric of the numerical vacuum curves under the pressure of the dimensional transition:

$$ds^2 = \tau(x)^2 dx^2 = \frac{\tau_0^2}{x} dx^2 \quad (38)$$

Integrating this curved metric yields the true, information-scaled physical distance $S(x)$ from the origin:

$$S(x) = \int_0^x \frac{\tau_0}{\sqrt{u}} du = 2\tau_0\sqrt{x} \quad (39)$$

Because the phase of the cosmic waveguide is directly coupled to this non-linear metric, any propagating electromagnetic wave experiences continuous geometric stretching of its wavelength. The cosmological redshift is therefore not a pure Doppler effect of physical recessional velocity, but an accumulated phase shift within the curved, self-reactive number space:

$$\Delta\Phi(x) = \int (\omega\tau(x) - k)dx \propto \sqrt{x} \quad (40)$$

When this geometric scaling is incorporated into cosmic distance measurements, the Hubble tension resolves completely without requiring the postulation of hypothetical Dark Energy.

6.2 The Cosmic Microwave Background (CMB) as the $D_3 \rightarrow D_4$ Event Horizon

Within this model, the Cosmic Microwave Background (CMB) is not the cooled remnant of a singular, initial Big Bang point-source. Instead, it is the continuous, isotropic blackbody radiation emitted due to thermodynamic friction and information loss directly at the global boundary membrane of the $D_3 \rightarrow D_4$ dimensional cascade.

Since our three-dimensional space (D_3) constitutes the holographic boundary of the expanding four-dimensional bulk space, this transition acts as a physical membrane. The thermal emission of this membrane corresponds precisely to the Hawking temperature of the global cosmic horizon.

6.3 Quantum Vacuum Fluctuations: Information Projection from D_4

The phenomenon of virtual particles and quantum vacuum fluctuations is modeled as the stochastic materialization of quantum information projected dimensionally from the over-arching D_4 bulk into our local spacetime.

In a D_4 structure, matter collapses into higher-dimensional black holes. According to the holographic principle and unitarity, this information is conserved; it is quantized at the event horizons, dimensionally reduced, and projected orthogonally to the D_4 plane into our D_3 spacetime. Quantum fluctuations are thus the local, three-dimensional intersection points of these incoming D_4 data packets.

6.4 Asymptotic Collapse: Cosmic Terminal State Scenario

The coupling of the cosmic vacuum to the fine-structure constant α dictates the dynamics of expansion. As expansion progresses, the system approaches the critical threshold $\alpha \rightarrow 1$. At this mathematical boundary, the metric of the vacuum collapses asymptotically, triggering a catastrophic, high-energy feedback loop. This feedback leads to the thermal detonation of the ancestral 4D black hole within whose event horizon our universe is embedded.

6.5 Dimensional Scaling of Fundamental Constants: G , $\hbar$, and the Cosmological Constant Λ

The embedding of our 3-dimensional observable universe (D_3) as a holographic membrane on the S^3 event horizon of a collapsing 4D star requires a systematic re-evaluation of what standard physics considers "fundamental constants." Rather than being static, arbitrary parameters, the Gravitational constant G , Planck's constant $\hbar$, and the Cosmological Constant Λ emerge as dynamical projection coefficients of the 4D mass accretion rate $\dot{M}_{4D}$ and the toroidal aspect ratio $\alpha = \frac{r}{2\pi R}$.

1. **The Dilution of Gravity (G_{3D}):** In our framework, gravity is not a fundamental 3D force but the projected leakage of the 4D bulk curvature onto the S^3 boundary. By applying Gauss's law across the dimensional interface, the effective 3D gravitational coupling G_{3D} scales inversely with the major toroidal radius $R(t)$:

$$G_{3D}(t) = \frac{G_{4D}}{2\pi R(t)} = \frac{G_{4D} \cdot \alpha(t)}{r(t)} \quad (41)$$

where G_{4D} is the static, higher-dimensional gravitational constant of the 4D bulk. This dynamically explains the hierarchy problem (the extreme weakness of gravity in D_3) as a geometric dilution effect: as the universe expands ($R(t) \rightarrow \infty$), gravity becomes progressively weaker.

2. **The Quantum of Action ($\hbar$) and Vacuum Thickness (r):** Planck's constant $\hbar$ represents the fundamental phase-action limit of the microscopic poloidal coordinate r . On the Riemannian number torus, the quantum of action is restricted by the minimum loop path of the toroidal vacuum sheet:

$$\hbar(t) = m_e \cdot c(t) \cdot r(t) = m_e \cdot f_0 \cdot \frac{r(t)^2}{\alpha(t)} \quad (42)$$

As the poloidal radius $r(t)$ remains anchored to the sub-unit quantum-vacuum interval while the major radius $R(t)$ relaxes expansionally, the conformal gauge coupling ensures that local quantum-mechanical measurements of $\hbar$ remain perfectly invariant for comoving observers, despite the underlying cosmic drift.

3. **Resolution of the Cosmological Constant Problem (Λ):** The "vacuum energy" or cosmological constant Λ is the source of the greatest discrepancy in modern physics (120 orders of magnitude). In this model, Λ is not an intrinsic zero-point energy density of flat space, but the physical density of the ongoing 4D mass influx $\dot{M}_{4D}$ distributed across the S^3 boundary volume $V_{3D} = 2\pi^2 R^3$:

$$\Lambda(t) \equiv 8\pi G_{3D} \rho_{acc} = \frac{3\dot{M}_{4D}}{R(t)^2 \cdot c(t)} \quad (43)$$

Since the major radius $R(t)$ expands asymptotically driven by 4D star collapse, the effective cosmological constant decays naturally as $\Lambda(t) \propto R(t)^{-2}$. This dynamic decay elegantly resolves the Coincidence Problem and explains why the observed Λ today is incredibly small, yet non-zero, without requiring any fine-tuning of Planck-scale vacuum fields.

7 Conclusion

The systematic restructuring of cosmology along the $D_0 \rightarrow D_4$ dimensional cascade resolves the mathematical and physical inconsistencies of the classical Λ CDM framework. By treating space, metrics, and physical constants as emergent phenomena of an underlying, geometric informational structure governed by the action-reaction principle (*actio = reactio*) against physical and informational state changes, we unify quantum effects, black hole thermodynamics, and cosmological scaling relations within a single, self-consistent, and singularity-free geometric framework.

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